

EMBEDDED HACKING

Reverse Engineering the Raspberry Pi Pico 2

Operation Black Start - Instructor Solution Key

ARM Cortex-M33 • GDB • Ghidra

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For educational and defensive security research only



The task and criterion headings in this key are word-for-word identical to `CTF-01-R.md`, so a student can match each criterion one-to-one.

Artifact Identity

Artifact	Value
Student image	<code>CTF-01.bin</code>
Flash image	<code>CTF-01.uf2</code>
Target	Raspberry Pi Pico 2 / RP2350 ARM Cortex-M33
Image base	<code>0x10000000</code>
Console	UART0, GPIO 0 TX / GPIO 1 RX, 115200 8N1

```
CTF-01.bin  425591AC17FF4C22206286EE6F40B06A89523483804850A41854A9B6F89D7B70
CTF-01.uf2  B1552EE3BB5763D96C65D8DF1C86984EE842EF1A823FDF5D94132B1E10FF9D16
```

Proof tool: `python3 scripts/verify_ctf.py` returns 11/11 checks passed against `CTF-01.bin`.

Task 1: Setup and Initial Analysis (15 points)

Solution

Criterion 1.1: Ghidra Project Setup (3 points). Import `CTF-01.bin` as Raw Binary, language ARM:LE:32:Cortex, base address `0x10000000`, then run auto-analysis.

Criterion 1.2: main() and Status-Loop Addresses (4 points).

Element	Address
<code>main()</code>	<code>0x100001E0</code>
Recurring status loop start	<code>0x10000234</code>

Element	Address
Loop back-edge (b.n 0x10000234)	0x10000266

The back-edge instruction at 0x10000266 is b.n 0x10000234 , encoded as bytes E5 E7 .

Criterion 1.3: Vector Table Decoding (4 points). First 32 bytes of CTF-01.bin :

```
00 20 08 20 5B 01 00 10 1B 01 00 10 1D 01 00 10
11 01 00 10 11 01 00 10 11 01 00 10 11 01 00 10
```

Evidence	Answer
Vector table base	0x10000000
Initial SP	0x20082000
Reset pointer (as stored)	0x1000015B
Reset instruction address	0x1000015A

Criterion 1.4: Thumb Addressing (4 points). The stored reset pointer 0x1000015B has bit 0 set to 1, which selects Thumb mode. Clearing bit 0 (0x1000015B & ~1) gives the real instruction address 0x1000015A . The equally odd interrupt vectors (0x1000011B , 0x1000011D , 0x10000111) are handled the same way.

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 1.1: Ghidra Project Setup	3	Correct project name, ARM:LE:32:Cortex (ARM Cortex 32-bit little endian), base 0x10000000	One item off	Not set up
Criterion 1.2: main() and Status-Loop Addresses	4	Both addresses correct	One correct	Neither found
Criterion 1.3: Vector Table Decoding	4	Correct base, initial SP, reset pointer	One missing	Not found
Criterion 1.4: Thumb Addressing	4	Correctly clears bit 0 and identifies main()	General explanation	Incorrect

Instructor Notes & Assembly

- Confirm the Ghidra import used Raw Binary , ARM:LE:32:Cortex , base 0x10000000 , and that auto-analysis completed before any address was read.
- Verify main() is 0x100001E0 , the loop head is 0x10000234 , and the back-edge bytes E5 E7 sit at 0x10000266 .
- The reset vector is stored as 0x1000015B ; the real Thumb entry is 0x1000015A . Do not accept the un-cleared 0x1000015B as an instruction address.

Task 2: Find and Patch Bug #1: The Miscalibrated Safety Threshold (30 points)

Solution

`grid_deviation` is `volatile`, so the compiler emits two independent compares: one for the operator status and one for the automated dispatch decision.

Criterion 2.1: Locate Compare Site A (5 points).

```
100001fc: 2b5e  cmp  r3, #94  @ 0x5e
```

Item	Value
Address	0x100001FC
File offset	0x1FC
Original bytes	5E 2B
Original instruction	cmp r3, #94

Criterion 2.2: Locate Compare Site B (5 points).

```
1000020a: 2b5e  cmp  r3, #94  @ 0x5e
```

Item	Value
Address	0x1000020A
File offset	0x20A
Original bytes	5E 2B
Original instruction	cmp r3, #94

Criterion 2.3: Correct Immediate-Value Reasoning (8 points). The source constant is `SAFE_THRESHOLD = 95` and the test is `x < 95`. For an unsigned value, `x < 95` is exactly `x <= 94`, which the compiler emits as `cmp r3, #94` plus `ite hi`. The correct engineering limit is `60`, so the test is `x < 60`, which is `x <= 59`. The correct patched immediate is therefore `0x3B (59)`, not `0x3C (60)`. Patching to `0x3C` would wrongly accept a reading of exactly 60.

Criterion 2.4: Patch Compare Site A (4 points).

Address	File Offset	Original	Patched	Before	After
0x100001FC	0x1FC	5E 2B	3B 2B	cmp r3, #94	cmp r3, #59

Criterion 2.5: Patch Compare Site B (4 points).

Address	File Offset	Original	Patched	Before	After
0x1000020A	0x20A	5E 2B	3B 2B	cmp r3, #94	cmp r3, #59

Criterion 2.6: Explain Why Both Sites Must Be Patched (4 points). `operator_state` (0x20000844) and `dispatch_state` (0x20000834) are each computed from their own read of `grid_deviation`. Patching only site A fixes the displayed text while the automated dispatch at site B still authorizes on the dangerous reading. Frozen reading `87 : 87 <= 94` is true (STABLE/AUTHORIZED, wrong); `87 <= 59` is false (CRITICAL/HELD, correct).

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 2.1: Locate Compare Site A	5	Correct address and original bytes	Address off	Not found
Criterion 2.2: Locate Compare Site B	5	Correct address and original bytes	Address off	Not found
Criterion 2.3: Correct Immediate-Value Reasoning	8	Explains the <code><</code> to <code><=</code> transform and gives 0x3B	Correct value, no reasoning	Wrong value
Criterion 2.4: Patch Compare Site A	4	Byte change verified	Wrong byte	Not patched
Criterion 2.5: Patch Compare Site B	4	Byte change verified	Wrong byte	Not patched
Criterion 2.6: Explain Why Both Sites Must Be Patched	4	Clear explanation of the two independent comparisons	Vague	Missing

Instructor Notes & Assembly

- Both sites must be patched: 0x100001FC for operator status and 0x1000020A for the automated dispatch decision.
- The correct immediate is 0x3B (59), not 0x3C (60); the source test is a strict `<`, compiled as `<= 94`.
- Verify the byte changes on hardware: after patching, `operator_state` (0x20000844) and `dispatch_state` (0x20000834) read 0 and 0.
- Ghidra ARM/Thumb Context Note:** In raw `.bin` files, patching an instruction that precedes an `IT` block (`ite hi`) using the GUI *Patch Instruction* action triggers Ghidra's `ReDisassembleCommand`. The re-disassembler encounters an internal context register conflict when trying to re-declare the `ITBlock` context over existing instructions, collapsing Thumb decoding into 32-bit ARM mode and swallowing Site B (0x1000020A). Students must patch using the Bytes window workflow (Clear `C` -> edit byte 5E -> 3B in Bytes window with pencil icon -> Disassemble `D`) to keep Site B visible and cleanly aligned.

Task 3: Find and Patch Bug #2: The False Signal Banner (20 points)

Solution

Criterion 3.1: Locate the Banner String (5 points).

String	Address
"SIGNAL: NORMAL\r"	0x10003678
"NORMAL" substring to patch	0x10003680

The string is loaded at call site 0x10000224 (ldr r0, [pc, #92] -> 0x10003678) and printed by bl __wrap_puts at 0x10000226. It is printed once at boot and never recomputed.

Criterion 3.2: Patch Six Characters (8 points). NORMAL and DANGER are both six ASCII characters, so the patch preserves the string length.

Address Range	Original Bytes	Patched Bytes
0x10003680 - 0x10003685	4E 4F 52 4D 41 4C	44 41 4E 47 45 52

Criterion 3.3: Character-by-Character Documentation (4 points).

Address	Original Char	Original Byte	Patched Char	Patched Byte
0x10003680	N	4E	D	44
0x10003681	O	4F	A	41
0x10003682	R	52	N	4E
0x10003683	M	4D	G	47
0x10003684	A	41	E	45
0x10003685	L	4C	R	52

Criterion 3.4: Explain the Danger of a Hardcoded Status Word (3 points). The banner never consults the reading, so it reports a healthy line even while the frozen reading is dangerous. Operators trust supervisory banners, so a hardcoded NORMAL masks the hazard and prevents manual intervention.

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 3.1: Locate the Banner String	5	Correct address	Approximate	Not found
Criterion 3.2: Patch Six Characters	8	All six bytes changed, length preserved	Correct text, wrong bytes documented	Wrong length
Criterion 3.3: Character-by-Character Documentation	4	Original vs patched byte for all six characters	Partial	Missing
Criterion 3.4: Explain the Danger of a Hardcoded Status Word	3	Clear, specific reasoning	Generic	Missing

Instructor Notes & Assembly

- NORMAL and DANGER are both six characters; the patch must not change the string length or overwrite adjacent flash.

- The banner is printed once at boot and never recomputed, so it is a separate defect from the threshold and must be graded independently.
- Confirm the patch covers `0x10003680` through `0x10003685` exactly.

Task 4: Recover the Quarantined Dispatch Frame (10 points)

Solution

Criterion 4.1: Recover the Dispatch Frame (6 points). Address `0x100037A0` in flash `.rodata`:

```
WORLDGRID:BLACKSTART:GRID-7:WATER-3
```

Criterion 4.2: Explain Why It Is Never Transmitted (4 points). `retain_dispatch_frame()` reads only the first character into a `volatile` local so the linker keeps the string, but the pointer is never passed to any print or UART routine. The frame is evidence only and must not be patched.

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 4.1: Recover the Dispatch Frame	6	Correct address and full text	Partial text	Not found
Criterion 4.2: Explain Why It Is Never Transmitted	4	Clear static-analysis explanation	Vague	Missing

Instructor Notes & Assembly

- Accept the full string `WORLDGRID:BLACKSTART:GRID-7:WATER-3` at `0x100037A0` in flash `.rodata`.
- The frame is evidence only. Penalize any submission that patches or rewrites it instead of recovering it.

Task 5: Export and Verify (20 points)

Solution

Criterion 5.1: Export CTF-01_fixed.bin (4 points). Export the patched program from Ghidra (`File -> Export Program...`, Binary Format) as `CTF-01_fixed.bin`.

Criterion 5.2: Convert to CTF-01_fixed.uf2 (4 points).

```
python uf2conv.py CTF-01_fixed.bin --base 0x10000000 --family 0xe48bff59 --output CTF-01_fixed.uf2
```

Criterion 5.3: Hardware Verification (8 points).

Before patching:

```
GLOBAL EMBEDDED RESPONSE NETWORK
BLACK START WINDOW: 27 MINUTES
UART0 115200 8N1 | AUTHORIZED LAB CONSOLE
SIGNAL: NORMAL
RESPONSE> GRID STATUS: STABLE
DISPATCH PATH: AUTHORIZED
LAST FRAME: QUARANTINED
RESPONSE>
```

After all three patches:

```
GLOBAL EMBEDDED RESPONSE NETWORK
BLACK START WINDOW: 27 MINUTES
UART0 115200 8N1 | AUTHORIZED LAB CONSOLE
SIGNAL: DANGER
RESPONSE> GRID STATUS: CRITICAL
DISPATCH PATH: HELD
LAST FRAME: QUARANTINED
RESPONSE>
```

If the console is not wired, the same proof is read over SWD: after reset, `operator_state ( 0x20000844 )` and `dispatch_state ( 0x20000834 )` are `1` and `1` for the shipped image, and `0` and `0` for the patched image. This has been confirmed on real hardware.

Criterion 5.4: Summary Table of All Patches (4 points).

#	What	Address(es)	Original	Patched
1a	Operator threshold	0x100001FC	5E 2B	3B 2B
1b	Dispatch threshold	0x1000020A	5E 2B	3B 2B
2	Signal banner	0x10003680 - 0x10003685	4E 4F 52 4D 41 4C	44 41 4E 47 45 52

Total: **8 bytes actually change** (one immediate byte at each compare site and six string bytes).

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 5.1: Export CTF-01_fixed.bin	4	Valid patched binary	Corrupted	Not submitted
Criterion 5.2: Convert to CTF-01_fixed.uf2	4	Correct base and family flags	Wrong flags	Not submitted
Criterion 5.3: Hardware Verification	8	Corrected console output confirmed on hardware	Some lines corrected	No verification
Criterion 5.4: Summary Table of All Patches	4	Complete address and before/after table	Missing entries	No table

Instructor Notes & Assembly

- Verify the exported image with `python3 scripts/verify_ctf.py`; the shipped check expects `11/11` checks passed against `CTF-01.bin`.

- Confirm the UF2 conversion used base `0x10000000` and family `0xe48bff59`.
- **Serial Terminal Timing:** Note that `print_boot_banner()` (`SIGNAL: DANGER`) fires within the first 5 milliseconds of boot. In normal lab usage, PuTTY attaches after boot and will display the continuous 1-second status loop (`GRID STATUS: CRITICAL` , `DISPATCH PATH: HELD`). To see `SIGNAL: DANGER` , the student must pulse `RUN` to `GND` while PuTTY is open, or demonstrate the string change at `0x10003680` via Ghidra static analysis.
- If no console is available, accept the SWD capture showing `operator_state` and `dispatch_state` at `0` and `0` on the patched image.

Task 6: Written Reflection (5 points)

Solution

Criterion 6.1: "Rushed Build" Is Not an Excuse (2 points). The missed review step is exactly what shipped the false-safe report; pressure explains why the safeguard was skipped, not why it should be skipped.

Criterion 6.2: One Engineering Practice per Bug (3 points).

- Bug #1 (duplicated threshold): a unit test or static-analysis rule that requires every comparison against `SAFE_THRESHOLD` to use one shared source of truth.
- Bug #2 (hardcoded banner): a hardware-in-the-loop smoke test that checks the boot banner against the latched reading.

Grading Rubric (1-to-1 Mapping)

Criterion	Points	Full Credit (Answer Key)	Partial Credit	No Credit
Criterion 6.1: "Rushed Build" Is Not an Excuse	2	Specific, grounded reasoning	Generic	Missing
Criterion 6.2: One Engineering Practice per Bug	3	Concrete practice for each bug	One bug only	Missing

Instructor Notes & Assembly

- Grade the specificity of the reasoning, not the length of the prose.
- Require one concrete engineering practice for each of the two bugs.

How To Breadboard

- **Raspberry Pi Pico 2** powered over USB.
- **3.3 V USB-UART adapter:**
 - Adapter RX to Pico GPo (UARTo TX)
 - Adapter TX to Pico GP1 (UARTo RX)
 - Adapter GND to Pico GND
 - Do not connect the adapter VCC while the Pico is USB powered.
- **Serial monitor:** 115200 baud, 8 data bits, no parity, 1 stop bit.
- No other peripherals are required; all evidence is obtained from the console.

Complete Grading Summary

Task	Title	Points
Task 1	Setup and Initial Analysis	15
Task 2	Find and Patch Bug #1: The Miscalibrated Safety Threshold	30
Task 3	Find and Patch Bug #2: The False Signal Banner	20
Task 4	Recover the Quarantined Dispatch Frame	10
Task 5	Export and Verify	20
Task 6	Written Reflection	5
TOTAL		100

Instructor Notes

Safety: Use only the supplied Pico 2, 3.3 V UART adapter, and firmware. Never connect the exercise to an operational grid, water plant, public network, military system, or third-party device.

Common Student Mistakes

- Patching only one threshold site (`0x100001FC` or `0x1000020A`), which leaves one status line lying.
- Assuming the immediate equals the limit, producing an off-by-one boundary; the correct byte is `0x3B` (59), not `0x3C` (60).
- Replacing the banner string with a different length, corrupting adjacent flash; `NORMAL` and `DANGER` are both 6 bytes.
- Treating the odd vector address `0x1000015B` as invalid instead of clearing bit 0 to get `0x1000015A`.
- Modifying the quarantined dispatch frame at `0x100037A0`, which destroys evidence.

Partial Credit Guidelines

- Award partial credit for one correct threshold site out of two, or for a correct immediate value without the `<` to `<=` reasoning.
- Award partial credit for a correct banner text with incorrectly documented bytes, or for partial character-by-character documentation.
- Accept an SWD read of `operator_state` and `dispatch_state` as equivalent proof when no UART console is available.
- Award no credit for patches that change string length or overwrite adjacent flash.

Appendix: Expected Binary Diff

#	What	File Offset(s)	Address(es)	Original Bytes	Patched Bytes
1a	Operator threshold	<code>0x1FC</code>	<code>0x100001FC</code>	<code>5E 2B</code>	<code>3B 2B</code>

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#	What	File Offset(s)	Address(es)	Original Bytes	Patched Bytes
1b	Dispatch threshold	0x20A	0x1000020A	5E 2B	3B 2B
2	Signal banner	0x3680 - 0x3685	0x10003680 - 0x10003685	4E 4F 52 4D 41 4C	44 41 4E 47 45 52

Total: **8 bytes actually change** (one immediate byte at each compare site and six string bytes).